Decoding Manchester Coded Transmissions in Hardware

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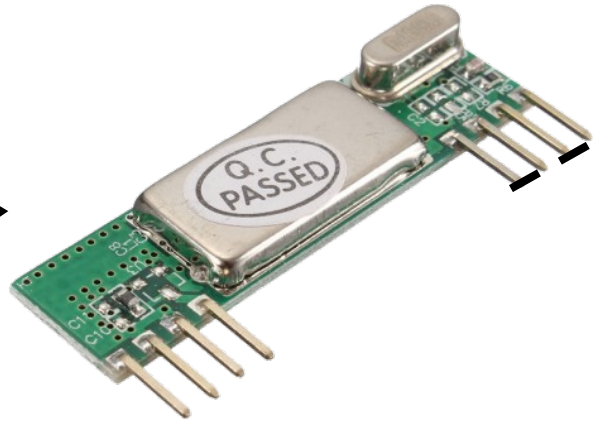
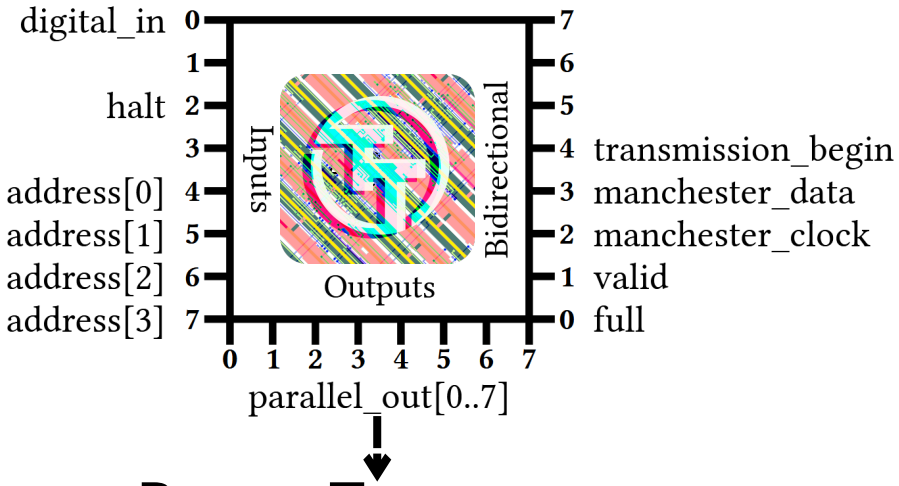
The Project

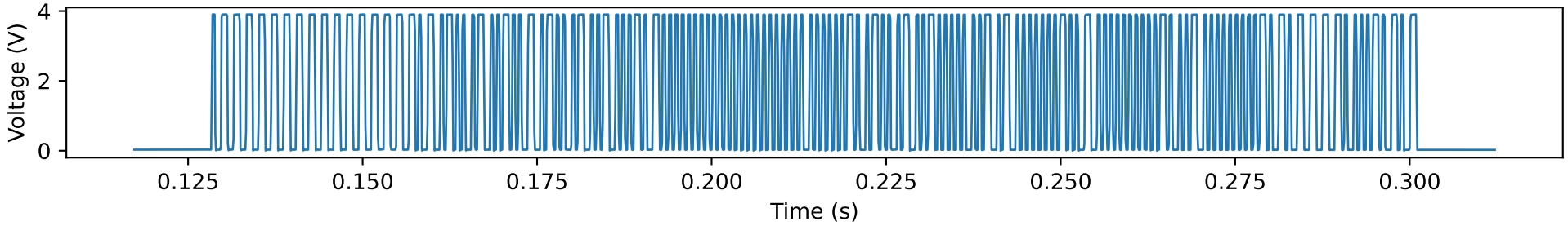
- Decoding Manchester-coded data from cheap devices like thermostats.
- Goal: Demonstrate an ASIC-based solution using **TinyTapeout**.
- Designed and simulated in 16 weeks (1 semester)



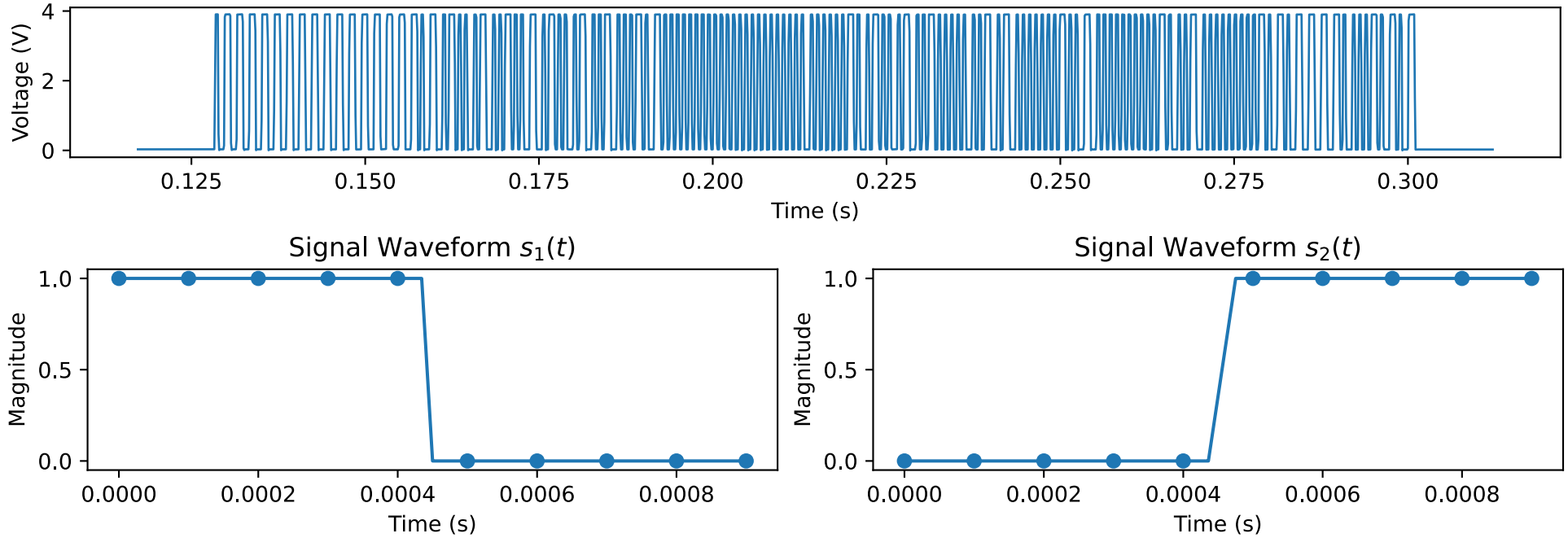


- Room Temp: 23.0°
- Set Temp: 18.0°

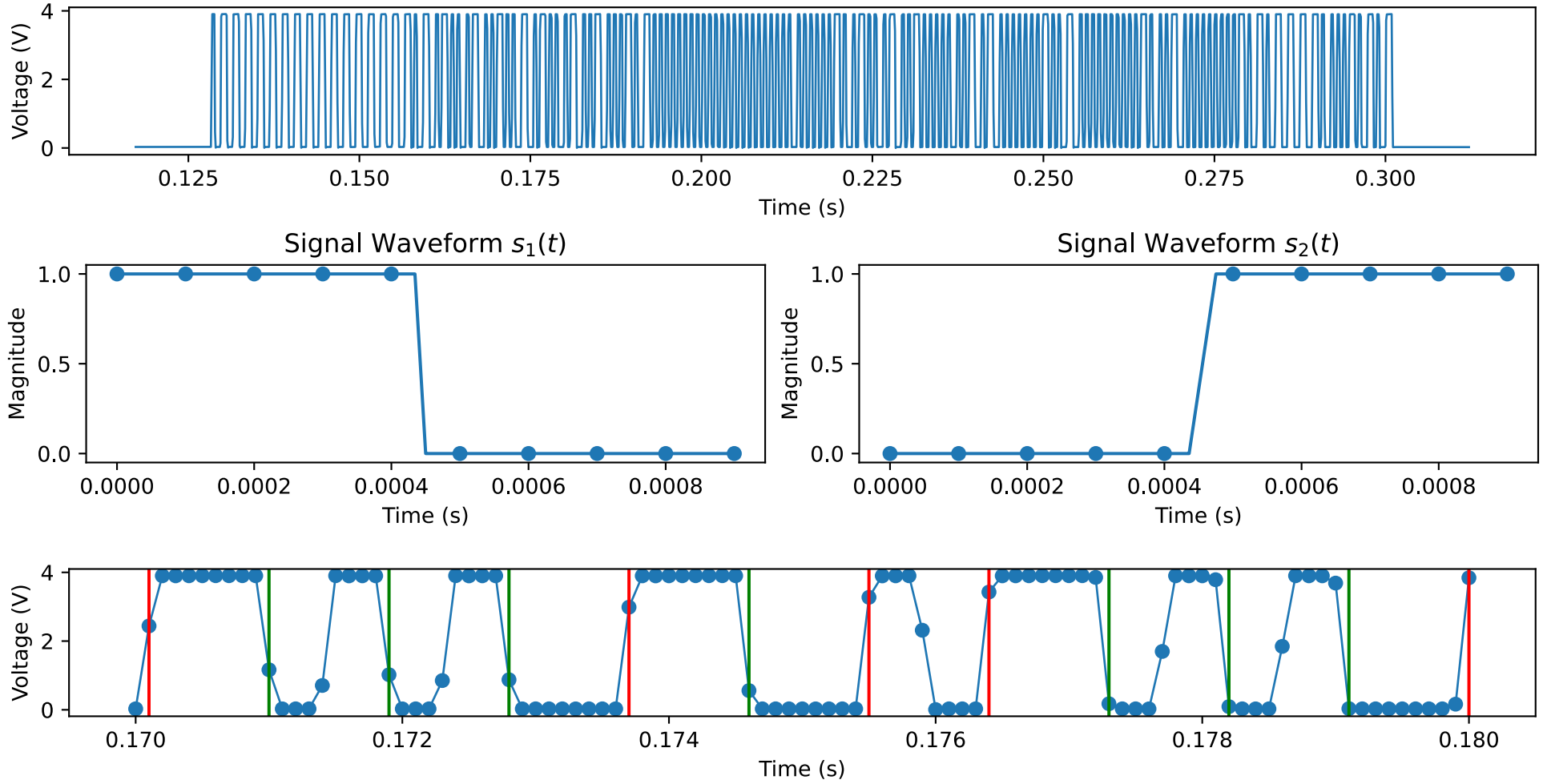
Manchester Coding



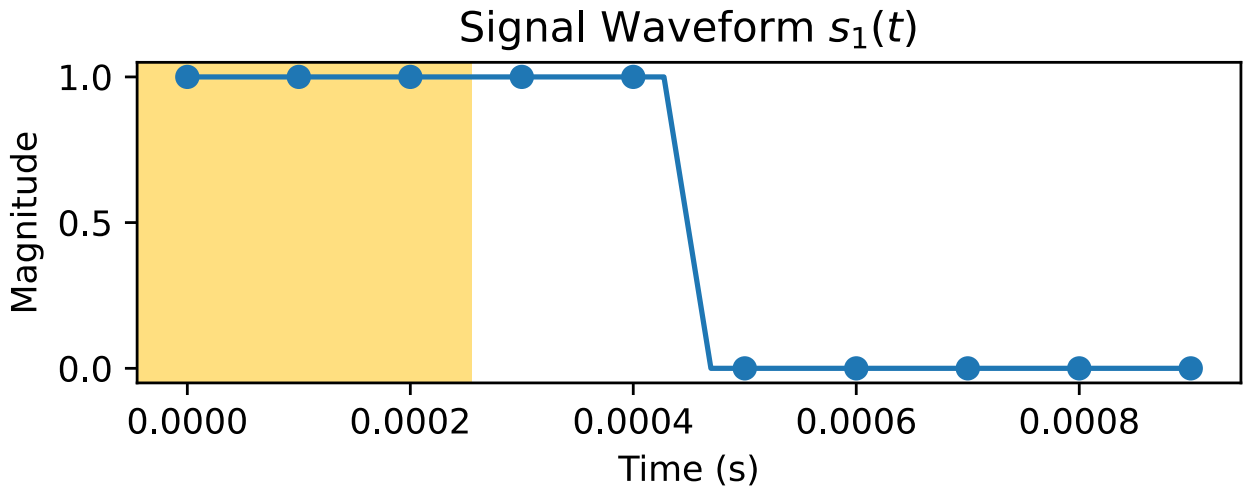
Manchester Coding

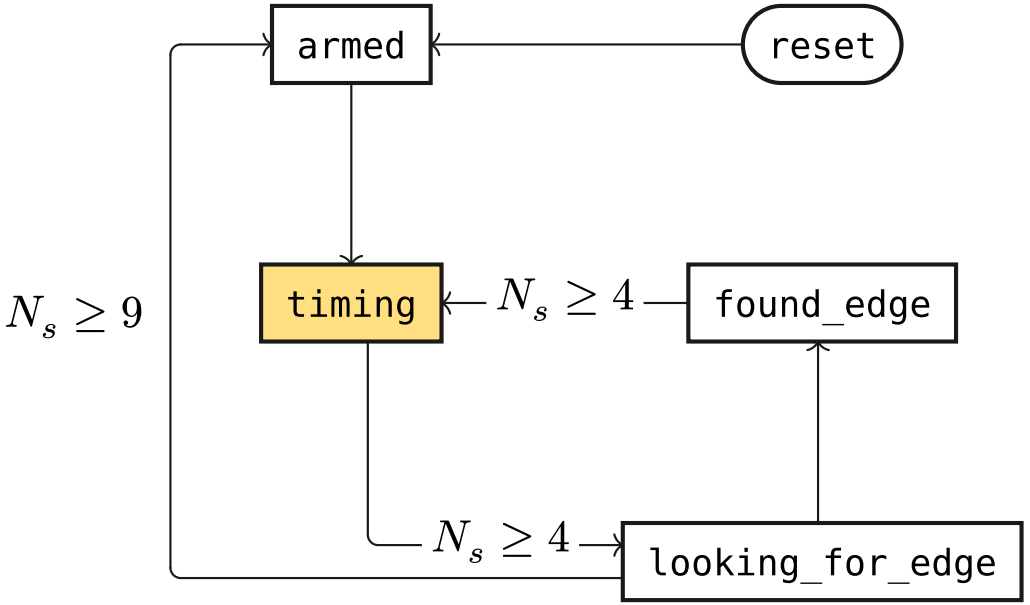


Manchester Coding

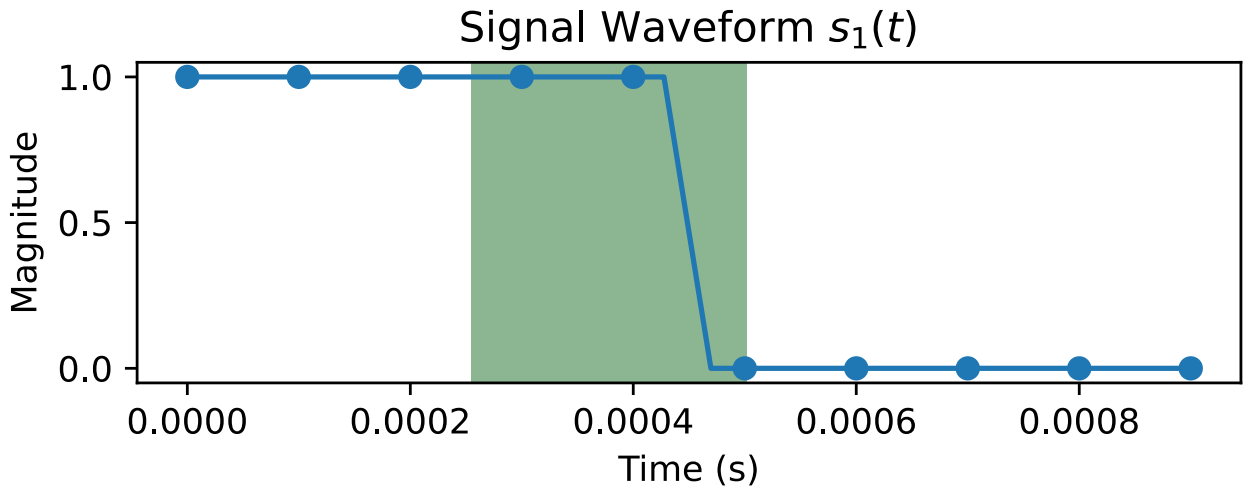


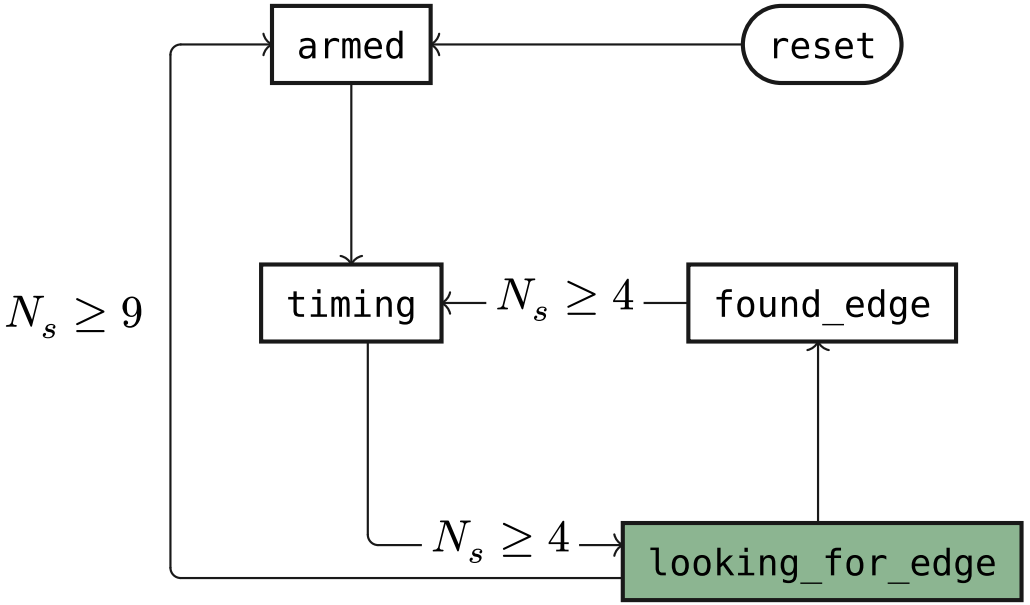
Decoder State Machine



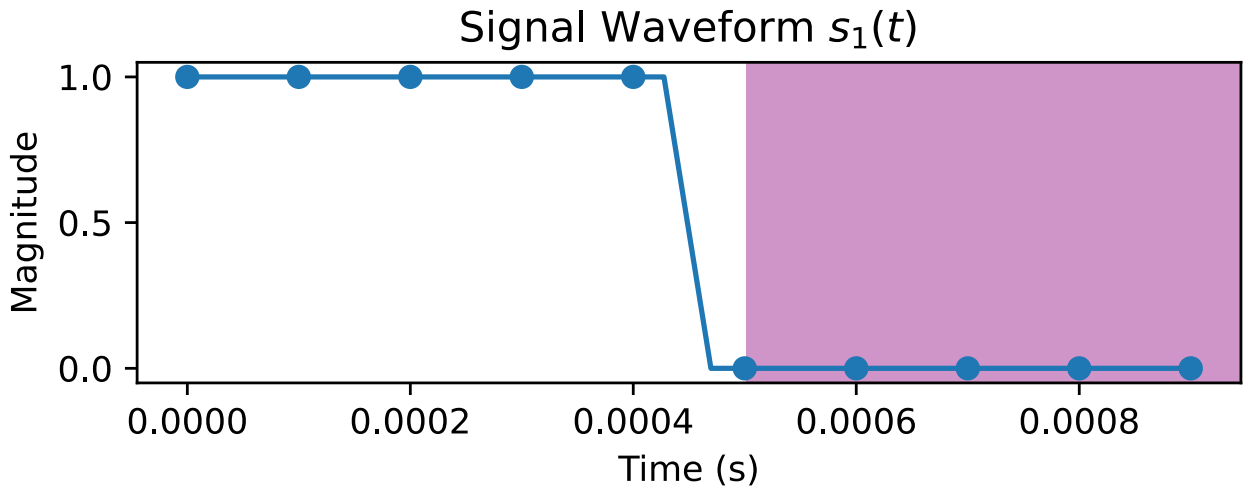


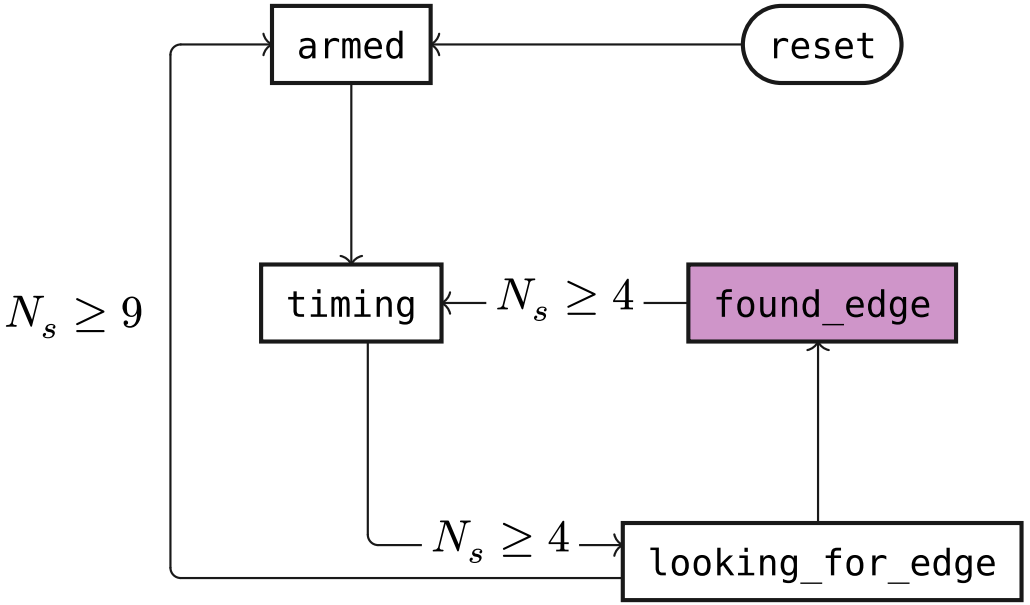
Decoder State Machine



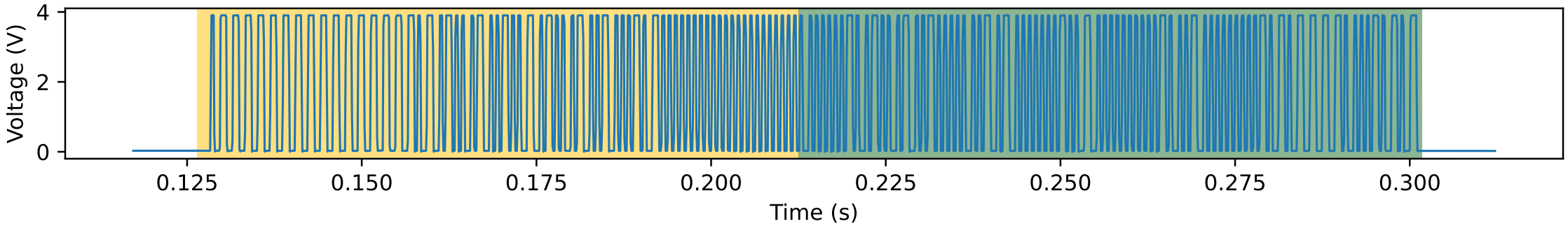


Decoder State Machine





Data Frame



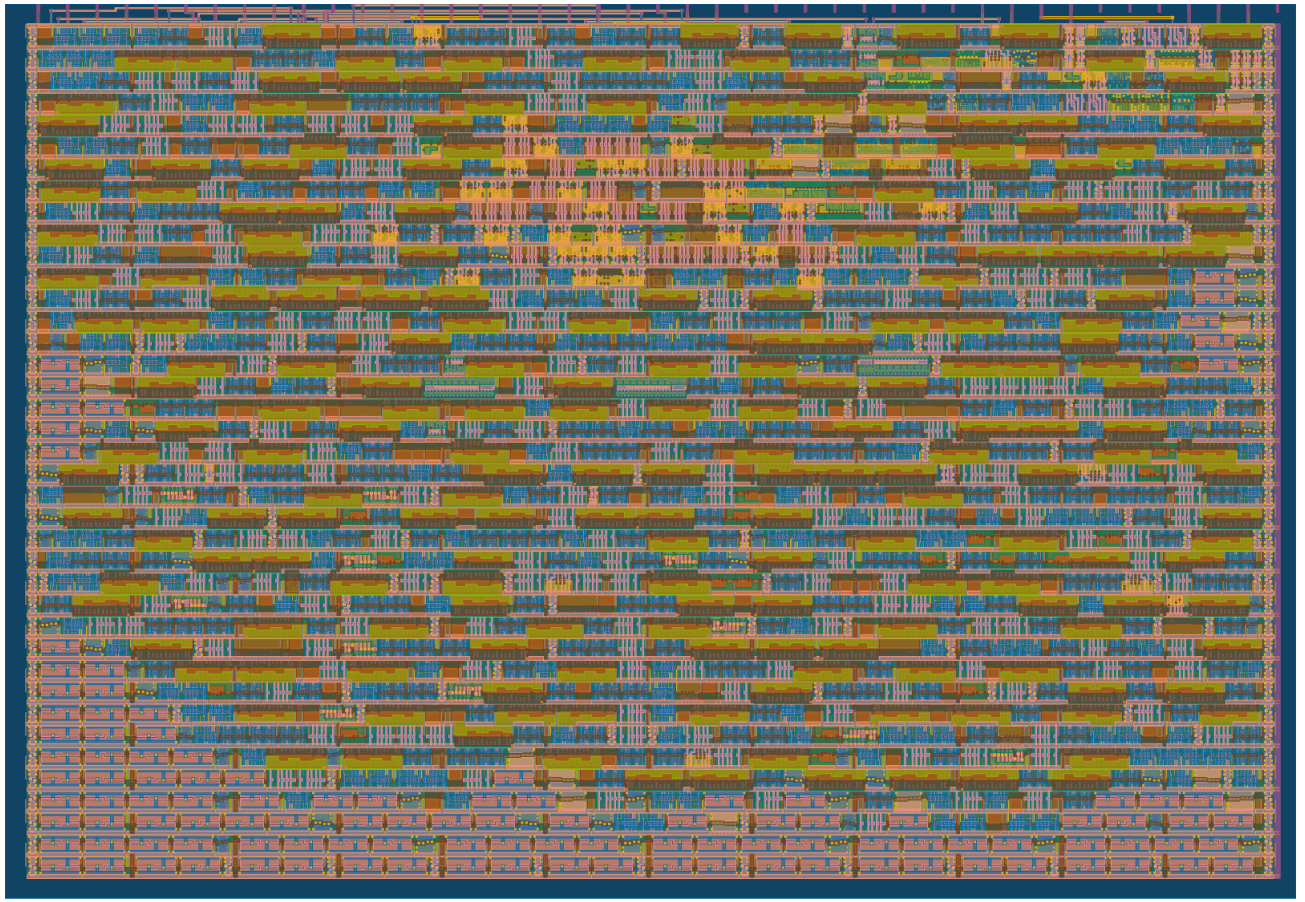
Preamble: Same for each transmission

Data: Variable, containing the interesting fields

Optimization

- Support from TinyTapeout Discord community

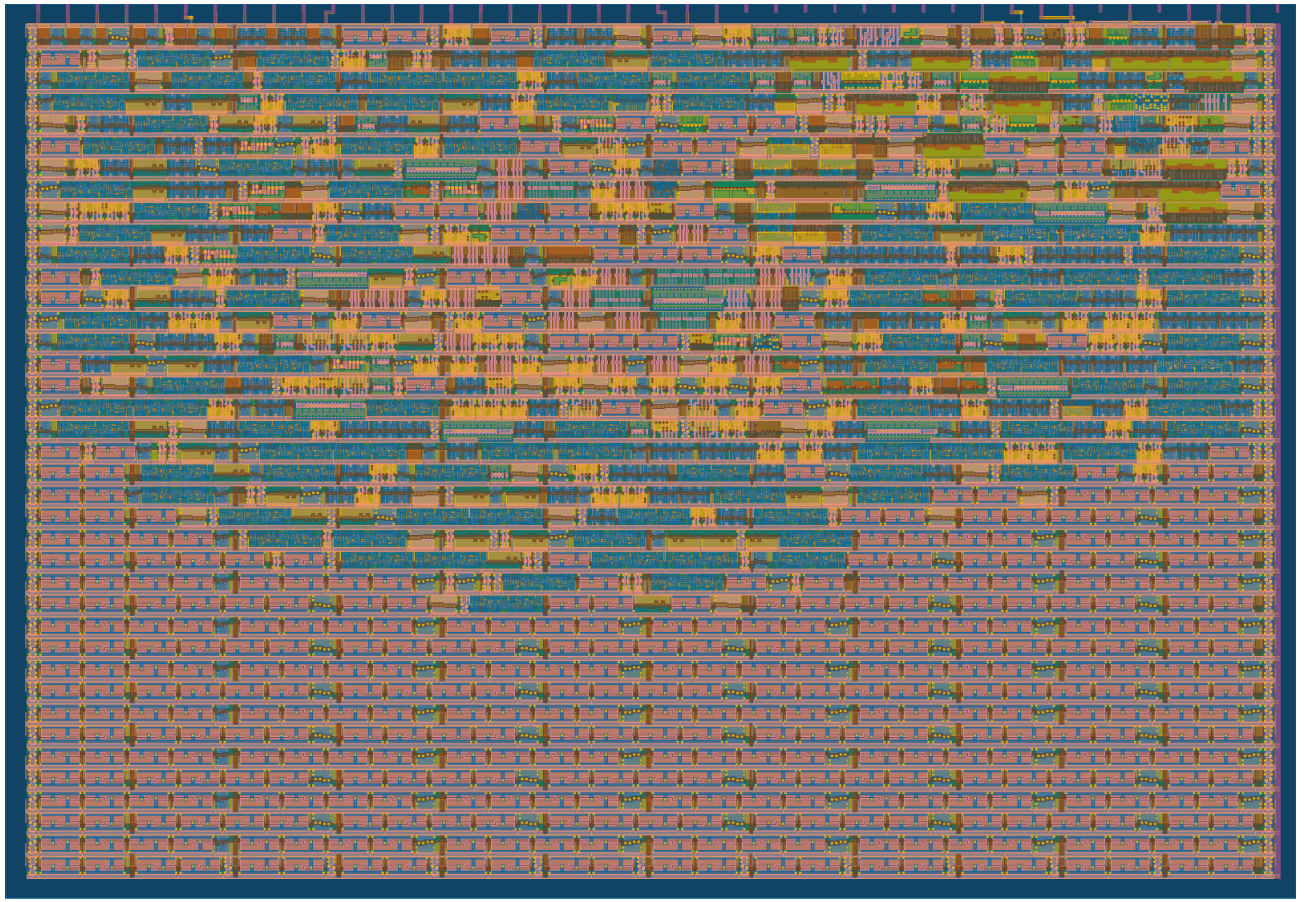
- ASIC fits within a single tile



Optimization

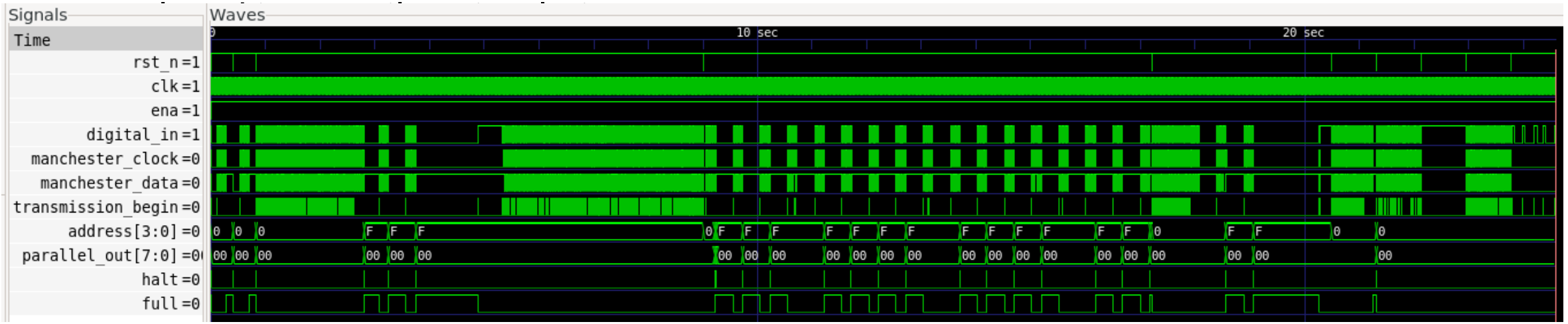
- Support from TinyTapeout Discord community

- ASIC fits within a single tile
- Optimized further from 62% to 32% utilization.



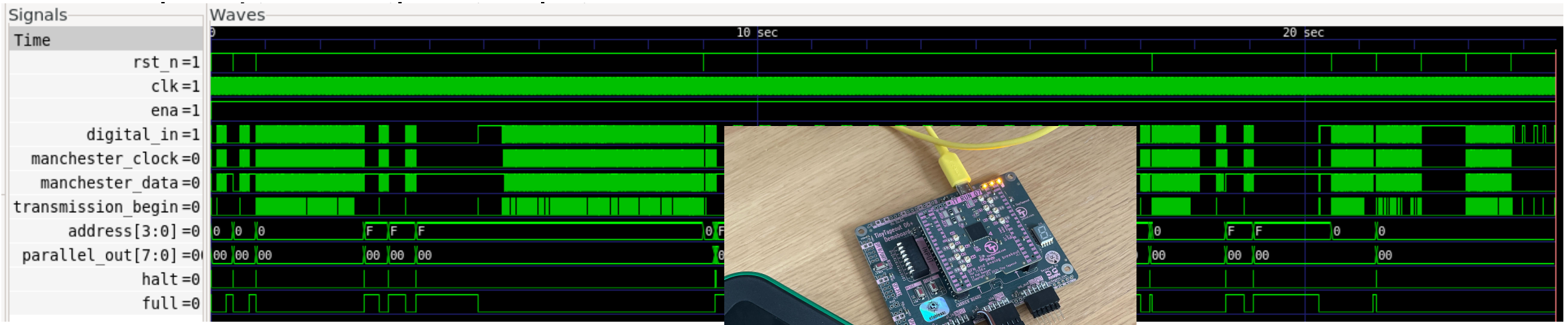
Simulation Testing

- Transmissions from the thermostat were captured with the RXB6



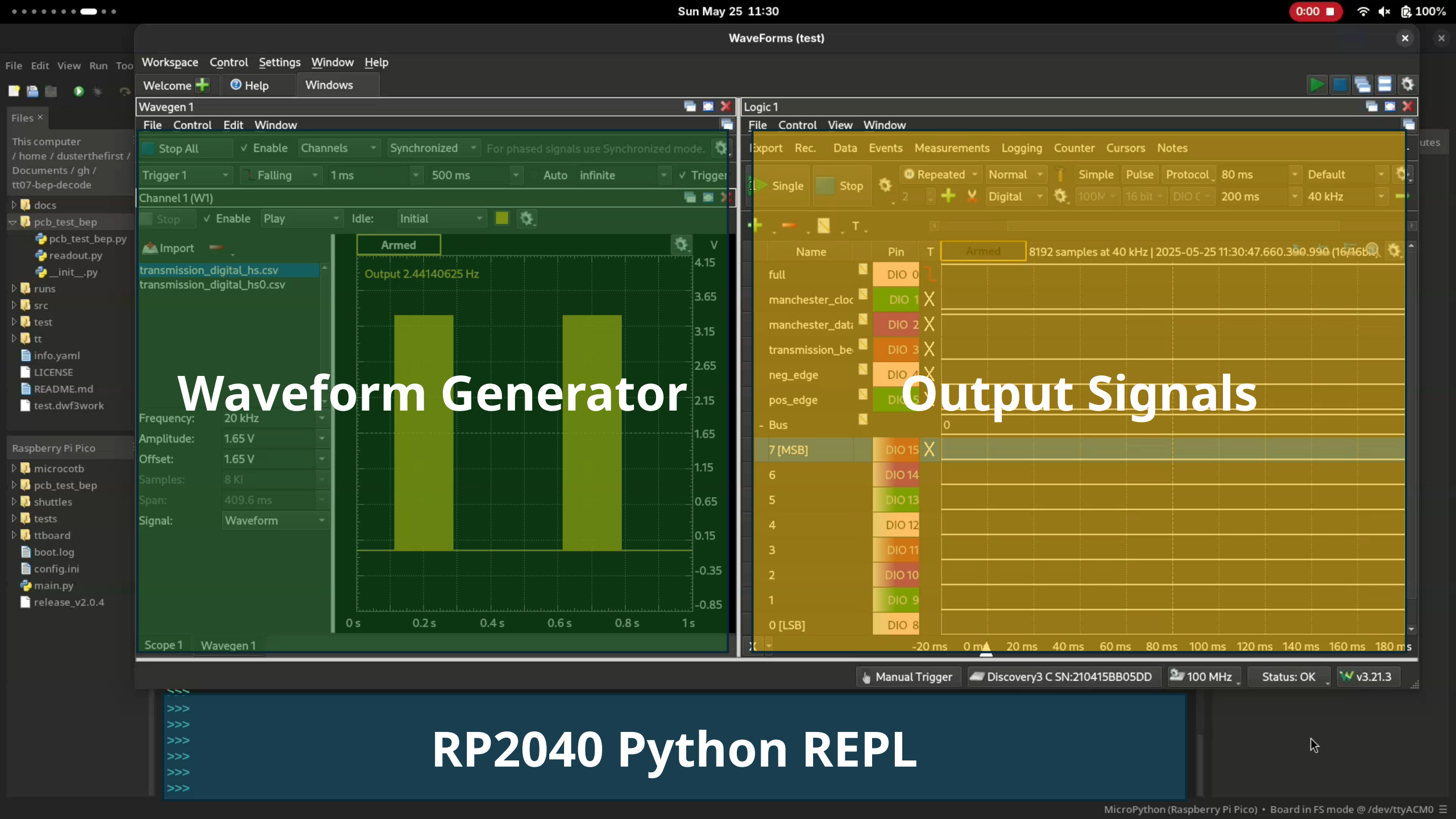
Simulation Testing

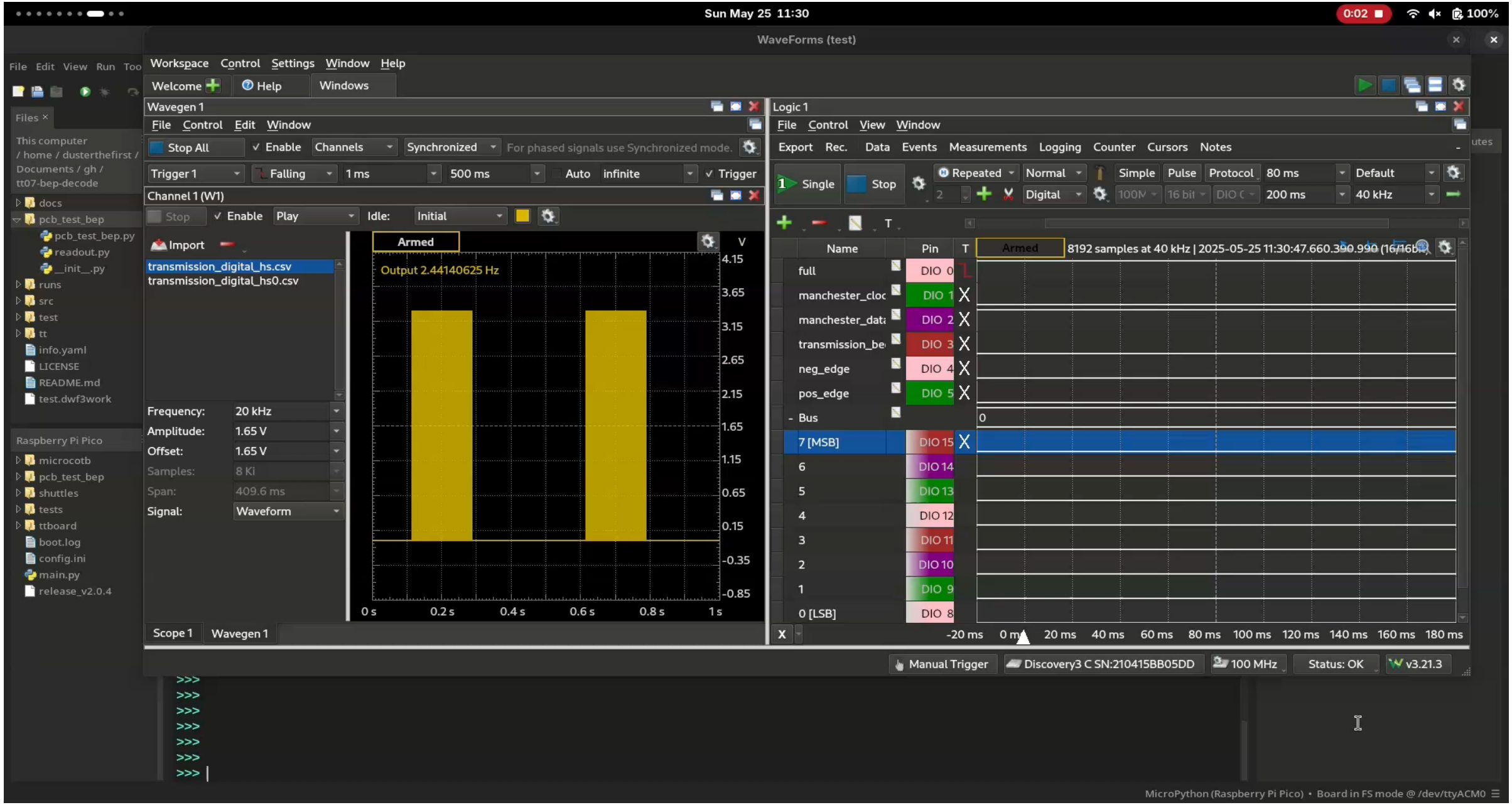
- Transmissions from the thermostat were captured with the RXB6

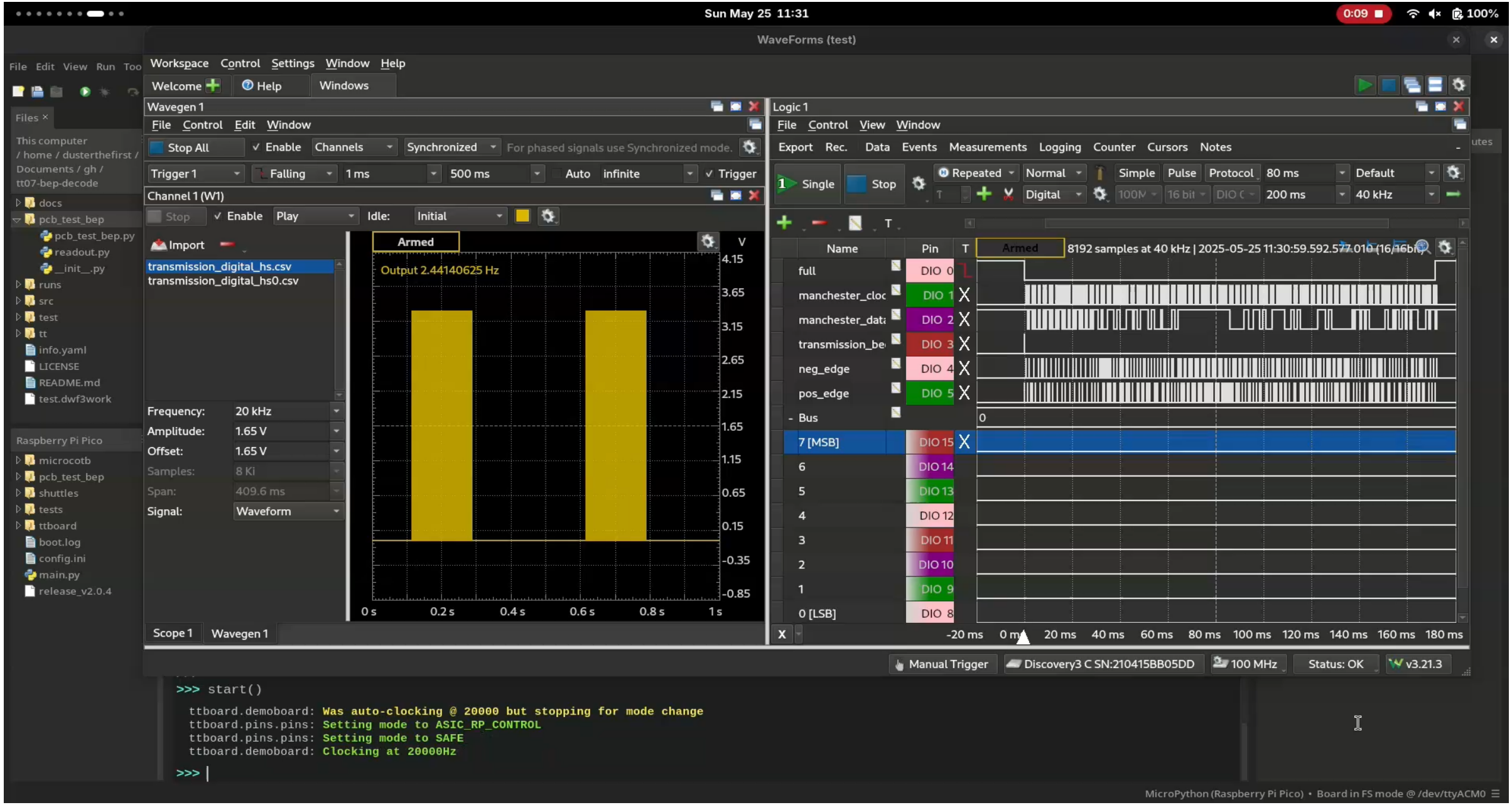


Real World Testing

- Tested with a logic analyzer and the onboard RP2040 microcontroller
  - Replayed captured signals

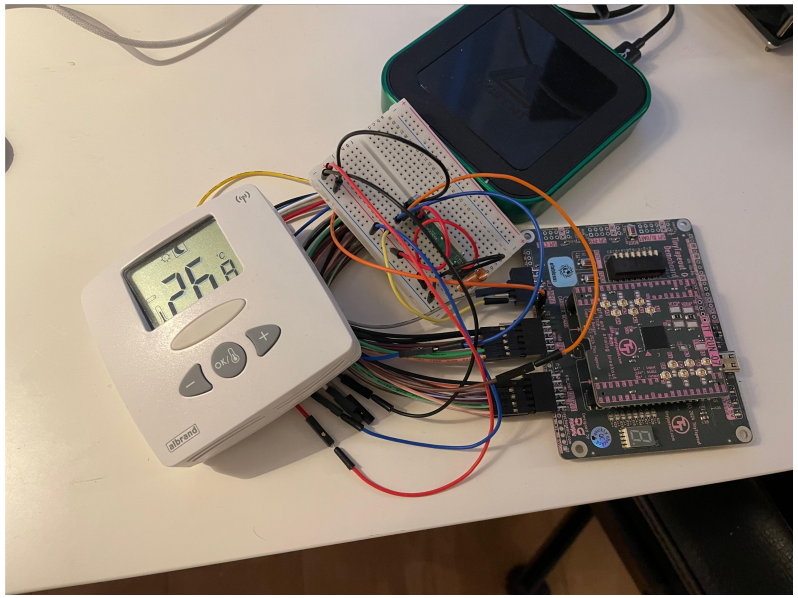

Sun May 25 11:30
WaveForms (test)
Workspace Control Settings Window Help
Welcome Help Windows
Wavegen 1
File Control Edit Window
Stop All Enable Channels Synchronized For phased signals use Synchronized mode.
Trigger 1 Falling 1 ms 500 ms Auto infinite Trigger
Channel 1 (W1)
Stop Enable Play Idle: Initial
Import
transmission_digital_hs.csv
transmission_digital_hs0.csv
Armed
Output 2.44140625 Hz
Frequency: 20 kHz
Amplitude: 1.65 V
Offset: 1.65 V
Samples: 8 Ki
Span: 409.6 ms
Signal: Waveform
Scope 1 Wavegen 1
Logic 1
File Control View Window
Export Rec. Data Events Measurements Logging Counter Cursors Notes
Single Stop Repeated Normal Simple Pulse Protocol 80 ms Default
Digital 200 ms 40 kHz
Name Pin
full DIO 0
manchester_cloc DIO 1
manchester_data DIO 2
transmission_be DIO 3
neg_edge DIO 4
pos_edge DIO 5
Bus
7 [MSB] DIO 15
6 DIO 14
5 DIO 13
4 DIO 12
3 DIO 11
2 DIO 10
1 DIO 9
0 [LSB] DIO 8
8192 samples at 40 kHz | 2025-05-25 11:30:47.660.390.990 (16/16b)
Manual Trigger
Discovery3 C SN:210415BB05DD
100 MHz
Status: OK
v3.21.3
Waveform Generator
Output Signals
RP2040 Python REPL
MicroPython (Raspberry Pi Pico) • Board in FS mode @ /dev/ttyACM0


Sun May 25 11:30
0:02
100%
WaveForms (test)
Workspace Control Settings Window Help
Welcome
Help
Windows
Wavegen 1
File Control Edit Window
Stop All
Enable
Channels
Synchronized
For phased signals use Synchronized mode.
Trigger 1
Falling
1 ms
500 ms
Auto
infinite
Trigger
Channel 1 (W1)
Stop
Enable
Play
Idle:
Initial
Import
transmission_digital_hs.csv
transmission_digital_hs0.csv
Armed
Output 2.44140625 Hz
V
4.15
3.65
3.15
2.65
2.15
1.65
1.15
0.65
0.15
-0.35
-0.85
0 s
0.2 s
0.4 s
0.6 s
0.8 s
1 s
Frequency: 20 kHz
Amplitude: 1.65 V
Offset: 1.65 V
Samples: 8 Ki
Span: 409.6 ms
Signal: Waveform
Scope 1
Wavegen 1
Logic 1
File Control View Window
Export Rec. Data Events Measurements Logging Counter Cursors Notes
Single
Stop
Repeated
Normal
Simple
Pulse
Protocol
80 ms
Default
Digital
200 ms
40 kHz
Name
Pin
T
Armed
8192 samples at 40 kHz | 2025-05-25 11:30:47.660.390.990 (16/16b)
full DIO 0
manchester_cloc DIO 1
manchester_data DIO 2
transmission_be DIO 3
neg_edge DIO 4
pos_edge DIO 5
Bus
7 [MSB] DIO 15
6 DIO 14
5 DIO 13
4 DIO 12
3 DIO 11
2 DIO 10
1 DIO 9
0 [LSB] DIO 8
-20 ms
0 ms
20 ms
40 ms
60 ms
80 ms
100 ms
120 ms
140 ms
160 ms
180 ms
Manual Trigger
Discovery3 C SN:210415BB05DD
100 MHz
Status: OK
v3.21.3
>>>
MicroPython (Raspberry Pi Pico) • Board in FS mode @ /dev/ttyACM0


Sun May 25 11:31
0:09
100%
WaveForms (test)
Workspace Control Settings Window Help
Welcome
Help
Windows
Wavegen 1
File Control Edit Window
Stop All
Enable
Channels
Synchronized
For phased signals use Synchronized mode.
Trigger 1
Falling
1 ms
500 ms
Auto
infinite
Trigger
Channel 1 (W1)
Stop
Enable
Play
Idle:
Initial
Import
transmission_digital_hs.csv
transmission_digital_hs0.csv
Armed
Output 2.44140625 Hz
V
4.15
3.65
3.15
2.65
2.15
1.65
1.15
0.65
0.15
-0.35
-0.85
0 s
0.2 s
0.4 s
0.6 s
0.8 s
1 s
Frequency: 20 kHz
Amplitude: 1.65 V
Offset: 1.65 V
Samples: 8 Ki
Span: 409.6 ms
Signal: Waveform
Scope 1
Wavegen 1
Logic 1
File Control View Window
Export Rec. Data Events Measurements Logging Counter Cursors Notes
Single
Stop
Repeated
Normal
Simple
Pulse
Protocol
80 ms
Default
Digital
200 ms
40 kHz
Name
Pin
T
Armed
8192 samples at 40 kHz | 2025-05-25 11:30:59.592.577.010 (16/16bit)
full DIO 0
manchester_cloc DIO 1
manchester_data DIO 2
transmission_be DIO 3
neg_edge DIO 4
pos_edge DIO 5
Bus
7 [MSB] DIO 15
6 DIO 14
5 DIO 13
4 DIO 12
3 DIO 11
2 DIO 10
1 DIO 9
0 [LSB] DIO 8
-20 ms
0 ms
20 ms
40 ms
60 ms
80 ms
100 ms
120 ms
140 ms
160 ms
180 ms
Manual Trigger
Discovery3 C SN:210415BB05DD
100 MHz
Status: OK
v3.21.3
>>> start()
ttboard.demoboard: Was auto-clocking @ 20000 but stopping for mode change
ttboard.pins.pins: Setting mode to ASIC_RP_CONTROL
ttboard.pins.pins: Setting mode to SAFE
ttboard.demoboard: Clocking at 20000Hz
>>>
MicroPython (Raspberry Pi Pico) • Board in FS mode @ /dev/ttyACM0

Current Work

- Real-time testing with the RXB6
- Live display of the thermostat’s data



Takeway

- Testing designs in simulation is important
- ASICs are accessible, even to those with no prior experience.
- https://github.com/DusterTheFirst/tt07-bep-decode

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